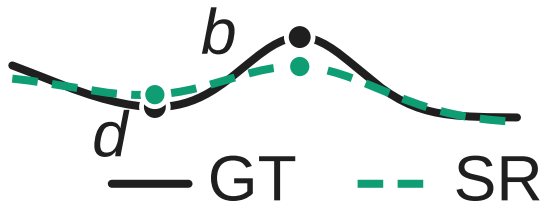


Critical-Value Supervision ($\mathcal{L}_{\text{TTK-CV}}$)

Matches predicted scalar values at fixed GT persistence-pair endpoints.



$$\mathcal{L}_{\text{TTK-CV}} = 0.5 [\text{MSE}(\hat{b}, b) + \text{MSE}(\hat{d}, d)]$$

b, d : GT endpoint values, $\hat{b}, \hat{d}$: SR values